

Vishal Kumar Allam

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PROFESSIONAL SUMMARY

- Forward Deployed Engineer and AI/ML professional with **6+ years of experience** delivering scalable software, data engineering, and AI solutions across enterprise environments, combining hands-on engineering with client-facing solution leadership and measurable business outcomes.
- Experienced developing **Generative AI, LLM-powered applications, Retrieval-Augmented Generation (RAG), autonomous agents, multi-agent orchestration, NLP, computer vision, and machine learning solutions** using Python and modern agent frameworks.
- Led solution lifecycles from **problem definition, technical discovery, architecture design, rapid prototyping, production-ready demos, deployment, scaling, and adoption**, collaborating directly with senior client stakeholders and multidisciplinary delivery teams.
- Strong hands-on expertise with **Python, LangChain, LlamaIndex, AutoGen, FastAPI, microservices, REST APIs, data platforms, big data technologies, PySpark, SQL, Kubernetes, Docker, and cloud-native architectures**.
- Experienced architecting and deploying solutions across **Azure, AWS, and GCP**, applying CI/CD, Infrastructure as Code with Terraform and Ansible, container orchestration, monitoring, logging, security, scalability, and resilient deployment patterns.
- Applied **MLOps, LLMops, AI governance, ethical AI frameworks, model evaluation, observability, model interpretability, and responsible AI practices** to improve reliability, transparency, compliance, and production readiness of enterprise AI solutions.
- Demonstrated emerging leadership through **client engagement, stakeholder management, mentoring engineers and data scientists, proposal development, technical demos, account-growth support, architecture reviews, and refinement of reusable FDE frameworks and best practices**.

EDUCATION

University of Missouri Kansas City

"Master of Science" in Computer Science

Kansas City, MO

Jan. 2023 – May 2024

Jawaharlal Nehru Technological University

"Bachelor of Technology" in Computer Science

Hyderabad, India

Aug. 2015 - May 2019

TECHNICAL SKILLS

- **Programming & Engineering:** Python, JavaScript, TypeScript, Java, C#, FastAPI, Flask, REST APIs, Microservices, Distributed Systems, Backend Engineering, Cloud-Native Development, System Design.
- **Generative AI & Agentic AI:** LLMs, RAG, Autonomous Agents, Multi-Agent Systems, Agentic Architectures, LangChain, LlamaIndex, AutoGen, Tool Calling, Prompt Engineering, Orchestration Frameworks, Cognitive Harness Patterns.
- **AI/ML & Data Science:** Machine Learning, NLP, Computer Vision, Predictive Modeling, Model Evaluation, Feature Engineering, PyTorch, TensorFlow, Scikit-learn, Embeddings, Vector Databases, Model Interpretability.
- **Data Engineering & Platforms:** PySpark, Spark, SQL, ETL/ELT, Data Pipelines, Data Modeling, Structured/Unstructured Data, Big Data, Databricks, Snowflake, Palantir Foundry/AIP, PostgreSQL, MySQL, SQL Server.
- **Cloud & Infrastructure:** Azure, AWS, GCP, Kubernetes, Docker, AWS Lambda, Azure Services, IAM, Terraform, Ansible, Infrastructure as Code, Container Orchestration, Cloud Security.
- **DevOps & AI Operations:** CI/CD, Azure DevOps, Jenkins, TeamCity, GitLab CI/CD, Git, MLOps, LLMops, Monitoring, Logging, Observability, Model Monitoring, AI Governance, Ethical AI, Responsible AI.
- **Consulting & Leadership:** Client Engagement, Technical Discovery, Solution Architecture, Rapid Prototyping, Production Demos, Proposal Development, Sales Pursuits, Account Growth, Stakeholder Management, Mentoring, Technical Leadership, Agile Delivery.

Software Engineer AI
PCMI LLC (Full Time)Oct 2024 – Present
Dallas, TX, USA

- Architected and delivered **AI-driven and data-centric enterprise solutions** using Palantir Foundry/AIP, Python, FastAPI, RAG, LLMs, autonomous agents, and cloud-native services, translating complex insurance business challenges into measurable technical outcomes.
- Led client and stakeholder discovery across the full solution lifecycle, including **problem definition, architecture design, rapid prototyping, production-ready demos, deployment planning, scaling, and adoption**, aligning AI technology roadmaps with operational and commercial objectives.
- Developed **LLM-powered applications and multi-agent systems** using LangChain, LlamaIndex, AutoGen, prompt engineering, tool calling, vector databases, semantic retrieval, and cognitive orchestration patterns to automate complex enterprise workflows.
- Engineered scalable AI and data platforms using **Python, PySpark, SQL, ETL/ELT, REST APIs, microservices, Palantir data pipelines, big data technologies, and structured/unstructured datasets**, supporting enterprise analytics and AI applications.
- Implemented cloud-native and production engineering practices across **AWS, Azure, GCP, Kubernetes, Docker, Terraform, Ansible, CI/CD, monitoring, logging, and Infrastructure as Code**, improving reliability, portability, and scalability across environments.
- Applied **MLOps, LLMops, AI governance, ethical AI, model evaluation, interpretability, observability, security, and responsible AI controls** to support safe and reliable deployment of AI/ML solutions within regulated insurance environments.
- Provided solution leadership through **technical demos, client presentations, architecture reviews, mentoring engineers and data scientists, proposal support, expansion opportunities, reusable FDE frameworks, and technical best practices** across cross-functional delivery pods.

Environments: Palantir Foundry/AIP, Python, FastAPI, LangChain, LlamaIndex, AutoGen, RAG, LLMs, Autonomous Agents, Multi-Agent Systems, PySpark, SQL, REST APIs, Microservices, AWS, Azure, GCP, Kubernetes, Docker, Terraform, Ansible, CI/CD, MLOps, LLMops, AI Governance, Observability.

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Software Engineer AI
Client: CVSDec 2023 – Sept 2024
Remote, USA

- Designed and delivered scalable healthcare AI and data applications using **Python, FastAPI, REST APIs, AWS, SQL, microservices, and cloud-native architecture**, supporting regulated enterprise workflows and data-intensive business processes.
- Developed **Generative AI, RAG, NLP, and agentic applications** using LLM APIs, LangChain, LlamaIndex, embeddings, vector databases, prompt engineering, and autonomous workflow patterns for healthcare information and operational use cases.
- Built enterprise data-processing solutions using **Python, PySpark, SQL, ETL/ELT, structured/unstructured datasets, APIs, and cloud data services**, enabling reliable information flow between analytics, AI, and operational platforms.
- Implemented scalable cloud and application architectures with **AWS, Docker, Kubernetes concepts, Terraform, CI/CD, REST APIs, OAuth2/OIDC, JWT, monitoring, and logging**, addressing reliability, security, privacy, and deployment requirements.
- Applied **MLOps/LLMops, model evaluation, observability, responsible AI, AI governance, performance monitoring, and model-output analysis** to improve reliability and production readiness of AI-enabled applications.
- Collaborated with product leaders, architects, ML engineers, data scientists, and business stakeholders through **technical discovery, solution design, architecture reviews, prototypes, production deployment, and adoption planning**.
- Supported technical leadership and business growth through **solution demonstrations, technical documentation, reusable components, mentoring, stakeholder communication, architecture trade-offs, and identification of opportunities to extend AI capabilities**.

Environments: Python, FastAPI, LangChain, LlamaIndex, RAG, NLP, LLM APIs, Vector Databases, PySpark, SQL, ETL/ELT, AWS, Docker, Kubernetes Concepts, Terraform, REST APIs, Microservices, CI/CD, MLOps, LLMops, AI Governance, Monitoring, Observability.

Software Engineer

Sept 2021 – Dec 2022

Infor

Hyderabad, India

- Designed and developed enterprise software and data platforms using **Python, FastAPI, Flask, REST APIs, SQL, microservices, and distributed-system architecture**, supporting scalable business applications and data-driven workflows.
- Built advanced data-processing pipelines using **Python, PySpark, Spark, SQL, ETL/ELT, data modeling, PostgreSQL, MySQL, Oracle, and big data technologies**, transforming high-volume datasets into actionable business information.
- Developed AI/ML capabilities involving **machine learning, NLP, predictive analytics, feature engineering, model integration, and data-driven decision support**, exposing outputs through reusable APIs and enterprise services.
- Integrated distributed applications using **REST APIs, microservices, event-driven architectures, authentication, asynchronous processing, and reusable service patterns**, improving interoperability across enterprise platforms.
- Applied cloud and DevOps practices using **AWS/Azure concepts, Docker, Kubernetes, Jenkins, Git, CI/CD, Terraform concepts, monitoring, and logging** to improve deployment consistency and application reliability.
- Collaborated with architects, product managers, data engineers, and business stakeholders to **analyze complex requirements, design scalable solutions, assess technical trade-offs, and align implementation decisions with broader business objectives**.
- Contributed to technical excellence through **architecture reviews, reusable engineering frameworks, code reviews, documentation, mentoring, knowledge sharing, and continuous improvement** across multidisciplinary engineering teams.

Environments: Python, FastAPI, Flask, PySpark, Spark, Machine Learning, NLP, REST APIs, Microservices, SQL, PostgreSQL, MySQL, Oracle, ETL/ELT, Big Data, Docker, Kubernetes, Jenkins, Git, CI/CD, Terraform Concepts, Monitoring, Cloud-Native Architecture.

Associate Engineer

June 2019 – August 2021

Virtusa

Hyderabad, India

- Developed enterprise backend and data-driven applications using **Python, Flask, REST APIs, SQL, and microservices architecture**, supporting complex operational workflows across distributed client environments.
- Built reusable Python components for **data processing, workflow orchestration, API integrations, business-rule execution, asynchronous services, and structured data transformation**, improving scalability and maintainability.
- Engineered enterprise data solutions using **Python, SQL, PostgreSQL, MySQL, batch pipelines, data modeling, ETL concepts, and high-volume transactional datasets**, enabling reliable downstream business applications.
- Designed distributed integrations across **REST APIs, microservices, external platforms, relational databases, authentication services, and event-driven workflows**, applying secure and resilient service-architecture patterns.
- Supported cloud-native and DevOps delivery through **Docker, Kubernetes concepts, Git, CI/CD, Linux, cloud infrastructure, structured logging, monitoring, and deployment automation**.
- Worked with architects, senior engineers, business analysts, and client stakeholders to **break down ambiguous requirements, evaluate technical options, develop implementation plans, and deliver scalable solutions aligned with operational priorities**.
- Created reusable **service components, API patterns, engineering documentation, architecture notes, mentoring materials, and knowledge-transfer assets**, strengthening technical standards and delivery capabilities across distributed teams.

Environments: Python, Flask, REST APIs, Microservices, SQL, PostgreSQL, MySQL, Data Modeling, ETL Concepts, Batch Processing, Event-Driven Architecture, Docker, Kubernetes Concepts, Git, CI/CD, Linux, Cloud Infrastructure, Logging, Monitoring, Distributed Systems.

AWARDS & ACHIEVEMENTS

- 2019-11 Oracle Certified Associate, JAVA SE8 Programmer.
- ISTQB Certified Professional.

- Microsoft Technology Associate (MTA) – Database Administration Fundamentals, HTML- 5 Application Development Fundamentals.
- Awarded “Software Process Achievement” Award for my contribution towards the project in Virtusa.
- Received Customer Appreciation mails for quality releases.